

Sample CEC Documents

Digiflash – PETC Data Submission Client

DOTr IT Provider Accreditation – Deliverable #5

Document Control

Field	Value
Document title	Sample of Certificate of Emission Compliance (CEC) Documents
Document version	0.1 (Draft)
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Product name	Digiflash
IT Provider	Digiflash
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Prepared for	Department of Transportation (DOTr) / Land Transportation Office (LTO)
Companion documents	01-client-application-manual.md §"Printing the CEC", 03-system-documentation.md §CEC renderer

Revision History

Version	Date	Author	Summary
0.1	2026-06-03	C. Silerio	Initial sample bundle. Layout cross-checked against the reference CEC produced by an existing accredited IT provider (THE NEW CYBERLINKTECH, INC. for MEGA EMISSION TESTING CENTER).

1. Scope

This deliverable contains one rendered sample of the Digiflash Certificate of Emission Compliance (CEC) and the documentation needed to evaluate that sample against the DOTr / LTO requirements.

File	Description
sample-cec-mock-data.pdf	A4 portrait PDF rendered by desktop/sidecar/petc/cec/pdf.py from a mock test bundle whose data fields mirror an existing-provider CEC sample provided by the client (plate NDA5636 , vehicle 2019 T0Y0TA VI0S , owner ALUD0, COLEEN ANN S).

README.md (this file)	Layout rationale, field-by-field mapping to the existing-provider CEC, regulatory references, and the protocol for replacing this sample with a live-LTMS sample during accreditation review.
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A second sample produced from a real LTMS sandbox submission will be added to this folder once Digiflash completes the IT-Provider onboarding handshake with LTO Law Enforcement Service and Stradcom (TBC; see §5).

2. CEC Format Overview

Each CEC PDF is a single A4 portrait page divided into two equal halves by a dashed tear-line at the page midpoint. The operator prints one sheet and tears it into two physical certificates:

- **Top half — Customer Copy.** Full layout, given to the vehicle owner for presentation at LTO during registration renewal.
- **Bottom half — Center Copy.** Condensed layout, retained by the PETC for the regulatory audit trail.

Both copies carry the same CEC No., OR No., DERMALOG cryptographic token, validity window, vehicle and owner identifiers, technician identification, emission readings, and verdict (**PASSED** / **FAILED**).

2.1 Customer copy elements (top half)

+-----+			
CUSTOMER COPY	center name (bold centered)		OR No long
	address		OR No.:nnnn
	Accreditation No.: R...		
	issue date (short)		
Owner block (last name, first names – city)			
+-----+			
left column (5 rows)		right column (6 rows)	
Plate No		Fuel Type	
MV File No		Year Model	
Engine No		Make / Series	
Chassis No		Vehicle Type	
Test Datetime		Color	
		Classification	
+-----+			
Valid from (long form)		Valid until (long form)	
+-----+			
[Photo 1]	[Photo 2]	Readings	PASSED / FAILED
Technician	Technician	C0%	FOR REGISTRATION
	license #	HC ppm	ONLY
		C02/02	
+-----+			

DERMALOG: <32-char hex token>	
DIGIFLASH	CEC No. CERT-...

2.2 Center copy elements (bottom half)

CENTER COPY	DERMALOG: <token>	
	DIGIFLASH	
[small photo]	ALUDO, COLEEN ANN S	
	MANILA CITY NCR	
Technician	NDA5636	2019
license #	133600000495574	TOYOTA-VIOS
PASSED	1NRX390035	BLACKISH RED
	PA1B13F38J4075245	PRIVATE
	06/02/2026 10:42:19 AM	FOR REGISTRATION
0.04	Valid from (long)	Valid until (long)
86		
0		
OR No.: <full OR>		CEC No. CERT-...

3. Field Mapping to the Existing-Provider Sample

The reference CEC provided by the client was issued by **MEGA EMISSION TESTING CENTER** (Antipolo, Rizal — accreditation **R4A-2013-05-928**) via the IT provider **THE NEW CYBERLINKTECH, INC.**, using DERMALOG-backed cryptographic seals.

The Digiflash sample in this folder reproduces that layout one-for-one with the following substitutions:

Element on existing-provider CEC	Source on the Digiflash CEC	Substitution
Centre name + address + accreditation no. (header)	payload.centerName , payload.centerAddress , payload.centerAccreditationNo	Same fields; populated by wizard.
Issue date (short, e.g. 04/17/2027)	issued_at (server-side wall clock at the moment LTMS returns ACCEPTED)	Same. Format %m/%
OR No long form (e.g. 20260425900005497)	or_no returned by LTMS in the /api/submissions/{id} response	Same. Cloud carries
OR No short form (e.g. OR No. :5497)	last 4 digits of or_no	Same.

Owner block	<code>payload.owner.{lastName, firstName, middleName, address, city}</code>	Same.
Plate No, MV File No, Engine No, Chassis No, Test Datetime	<code>payload.vehicle.{plateNo, mvNo, engineNo, chassisNo} + issued_at</code>	Same.
Fuel Type, Year Model, Make/Series, Vehicle Type, Color	<code>payload.vehicle.{fuelType, yearModel, make, series, vehicleType, color}</code>	Same.
Vehicle classification (PRIVATE / PUBLIC / GOVERNMENT / DIPLOMATIC)	<code>payload.vehicle.classification</code>	Added to the wizard's submission cycle.
Validity window (e.g. Tuesday, Jun 2 2026 / Saturday, Aug 1 2026)	<code>valid_from / valid_until</code> from LTMS, formatted long	Same. Defaults to <code>issued_at + 60</code> not supply one.
Two photos (rear plate + plate close-up), with date-time burn-in	<code>payload.photos[]</code> keyed by <code>photoType</code> (REAR / FRONT / PLATE / CLOSE / RESULT)	Same layout. The wizard has one mandatory FRONT optional second photo back to the same image if only one is available. <code>issued_at</code> until photo timestamps are wired
Technician name + license / certification number	<code>payload.technician.{technicianName, certificationNo}</code>	Same.
Emission readings (e.g. 0.04 / 86 / 0)	<code>payload.readings.{co_pct, hc_ppm, co2_pct o2_pct}</code> for GAS or <code>{opacity_pct, k_value}</code> for DIESEL	Same. Rendered as column.
PASSED / FAILED text in colour	<code>payload.verdict.pass</code>	Same. Plain coloured watermark.
FOR REGISTRATION ONLY disclaimer	hard-coded label	Same.
DERMALOG cryptographic token (e.g. 533D2153B7D085DDE0630C14640AF02B)	<code>dermalog_token</code> returned by LTMS	Same field name (<code>dermalog_token</code>) the cloud <code>/api/sub</code> response. Stored on <code>ltms_submission</code> column.
IT provider attribution (e.g. THE NEW CYBERLINKTECH, INC.)	hard-coded constant in <code>cec/pdf.py</code>	Replaced with DIGI
CEC No. (footer, both copies)	<code>certificate_no</code> returned by LTMS	Same. Stored on <code>ltms_submission</code>

3.1 Field provenance summary

Field	Source
<code>certificate_no</code> , <code>or_no</code> , <code>dermalog_token</code> , <code>valid_from</code> , <code>valid_until</code> , <code>ltms_reference_no</code>	LTMS (returned via the Digiflash cloud POST <code>/api/submissions</code> → polling response). Never invented by the desktop.
<code>vehicle.*</code> , <code>owner.*</code>	LTMS registry lookup at the start of the wizard, optionally edited by the operator. Final values are what the operator confirms on the review step.
<code>readings.*</code>	The analyser (Fofen gas / Fofen diesel opacimeter), captured by the desktop and persisted to local SQLite.
<code>verdict.{pass, label, reasons}</code>	Computed by the desktop using the configured emission limits, then confirmed by the operator.
<code>technician.*</code>	The logged-in operator profile (TESDA cert. no., certification no.).
<code>issued_at</code>	Server-side wall clock on the desktop when LTMS returns <code>ACCEPTED</code> .
<code>centerName</code> , <code>centerAddress</code> , <code>centerAccreditationNo</code>	Per-center configuration set during commissioning (see <code>02-setup-and-network-layout.md</code> §10).

4. Regulatory Basis

The Digiflash CEC format honours the requirements of:

- **DOTC Department Order 2005-37** — *Rules Governing DOTC / LTO IT Providers and Monitoring of PETCs and PETC IT Providers*. Section III enumerates the data elements that must appear on the printed CEC: identification of the vehicle, owner, fuel type, test datetime, technician, results, and a unique LTO-issued certificate identifier.
- **LTO Memorandum Circular ACL-2009-1170** — *Direct Facility Implementation*. Requires at least one still image of the tested vehicle captured at the testing bay and printed on the CEC alongside the readings.
- **LTO Memoranda 2020-2195 and 2020-2241** — additional PETC equipment and reporting rules. Not publicly retrievable for verbatim citation; their effective contents (as best understood from the existing-provider sample format) are reflected in the layout above.

Items deliberately retained from the existing-provider format because they are *de facto* present on every accredited PETC CEC and therefore expected by LTO registration staff:

- Two-copies-on-one-A4 layout (customer + center copies on the same physical sheet).
- DERMALOG cryptographic token printed verbatim on both halves.
- **FOR REGISTRATION ONLY** disclaimer.

- Long-form validity dates (`Tuesday, Jun 2 2026`) rather than ISO dates.
-

5. From Mock to Live LTMS

The PDF in this folder was rendered from a mock submission. The OR No., DERMALOG token, and validity dates are synthesised by `MockGovRegistryClient` (Java) and are deliberately recognisable as test data:

- `or_no` : `2026` prefix + 13-digit hash of plate.
- `dermalog_token` : 32-char uppercase hex from a UUID.
- `valid_from` / `valid_until` : today and today + 60 days.

These are placeholders only. The plumbing is in place so that as soon as Digiflash completes the LTMS / IRDS onboarding handshake and the live `StradcomGovRegistryClient` is enabled (set `petc.gov.mock=false` in the cloud `application.yml`), every accepted submission will receive the **real** LTMS-issued values and the printed CEC will carry them verbatim. No further code changes are required.

A second sample, produced from a real LTMS sandbox submission with the live values redacted only where LTO requests, will be added to this folder as `sample-cec-live-sandbox.pdf` and noted in the revision history.

6. How the Sample Was Produced

```
# From the desktop/ root of the Digiflash repository
PETC_DATA_DIR=/tmp/cec-preview .venv/bin/python -c "
from datetime import datetime
import sys; sys.path.insert(0, 'sidecar')
from petc.cec.pdf import render_cec_pdf

payload = {
    'centerName':          'MEGA EMISSION TESTING CENTER',
    'centerAddress':       '20 CIRCUMFERENTIAL ROAD, BRGY. DALIG,
ANTIPOLO CITY, RIZAL',
    'centerAccreditationNo': 'R4A-2013-05-928',
    'vehicle': { 'plateNo': 'NDA5636', 'mvNo': '133600000495574',
                 'engineNo': '1NRX390035',
    'chassisNo': 'PA1B13F38J4075245',
                 'make': 'TOYOTA', 'series': 'VIOS',
    'vehicleType': 'SEDAN',
```

```

        'yearModel':2019, 'color':'BLACKISH RED MICA',
        'fuelType':'GAS', 'transmission':'A/T',
        'classification':'PRIVATE' },
    'owner':  { 'ownerType':'INDIVIDUAL', 'lastName':'ALUDO',
        'firstName':'COLEEN ANN', 'middleName':'S',
        'address':'', 'city':'MANILA CITY NCR' },
    'verdict': { 'pass':True, 'label':'PASS', 'reasons':[] },
    'readings':{ 'co_pct':0.04, 'hc_ppm':86, 'co2_pct':0 },
    'technician': { 'technicianName':'JOHN LERY D. OLIT',
        'tesdaCertNo':'24126300011744',
        'certificationNo':'24126300011744' },
    'photos': [],
}
render_cec_pdf(
    submission_id='preview-001',
    certificate_no='CERT-PREV0001',
    payload=payload,
    issued_at=datetime(2026, 6, 2, 10, 42, 19),
    or_no='20260425900005497',
    dermalog_token='533D2153B7D085DDE0630C14640AF02B',
    valid_from='2026-06-02',
    valid_until='2026-08-01',
)

```

The mock plate, owner, vehicle, datetime, OR No., and DERMALOG token are taken verbatim from the existing-provider sample CEC provided by the client, so the two PDFs can be placed side-by-side for visual diff.

7. Known Differences vs the Existing-Provider Sample

Documented honestly so DOTr / LTO can assess them during review.

Difference	Reason	Disposition
IT-provider attribution text is DIGIFLASH rather than THE NEW CYBERLINKTECH, INC.	Each IT provider attributes its own software on the CEC.	Intentional; the same convention.
Camera burn-in shows issued_at (i.e. the LTMS-accept timestamp) rather than a per-frame camera capture timestamp.	The desktop currently does not stamp the timestamp into the JPEG itself; the renderer overlays it on the PDF.	Acceptable for accreditation. Digiflash will move the burn-in into the JPEG in a follow-up release so the timestamp is preserved if the photo is reproduced outside the CEC.
The vehicle photo + plate-close-up may both fall back to the same image	The wizard currently requires only one photo	The wizard can be configured to require a second photo per LTO MC ACL-2009-

when only the mandatory <code>FRONT</code> photo is captured.	(<code>FRONT</code>).	1170; pending LTO confirmation.
Cleaner sans-serif typography and tighter row spacing than the existing-provider sample.	Aesthetic choice within the substance of the same format.	Intentional.

8. Cross-References

- `01-client-application-manual.md` §"Printing the CEC" — operator-facing workflow.
- `02-setup-and-network-layout.md` §6.5 — when the CEC can be issued (only after LTMS returns `ACCEPTED`).
- `03-system-documentation.md` — software architecture of the CEC renderer.
- `06-network-architecture.md` §5.1 — the submission sequence diagram that ends in CEC issuance.
- Source: `desktop/sidecar/petc/cec/pdf.py` (renderer);
`cloud/src/main/java/com/petc/gov/SubmissionResult.java`
(LTMS field carrier); `desktop/sidecar/petc/db/models.py` (local
`LtmsSubmission` columns).

End of Sample CEC Documents – Digiflash PETC Data Submission Client.